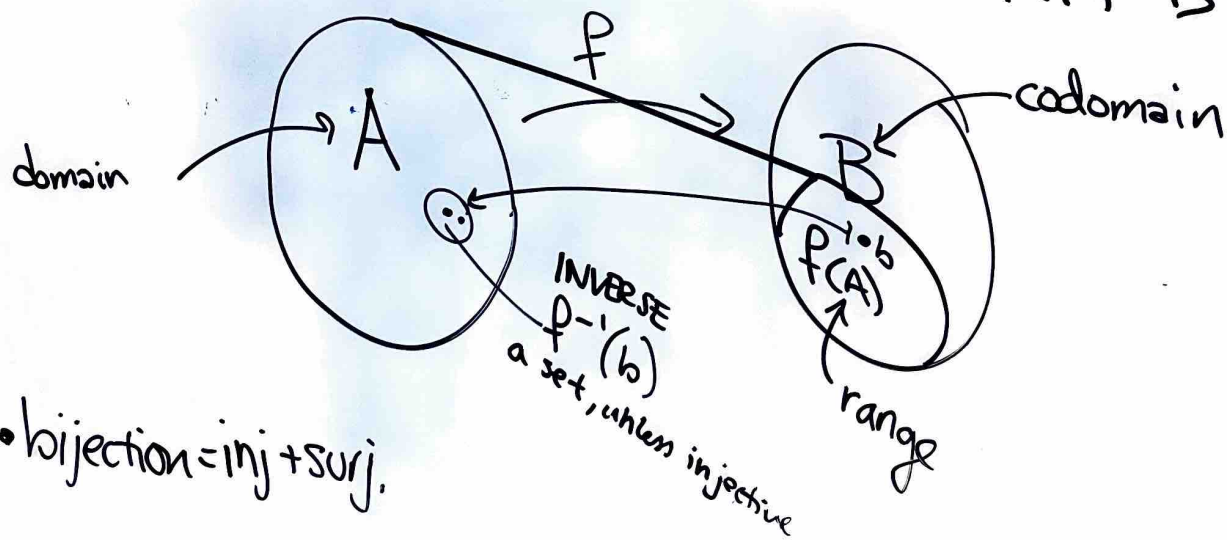


A fn  $f: A \rightarrow B$  is  
 a relation btwn  $A$  &  $B$

s.t. each elt. of  $A$  appears exactly once.

• injection:  $f(x) = f(y) \Rightarrow x = y$   
 (1-1)

• surjection:  $\forall b \in B, \exists a \in A$  s.t.  $f(a) = b$   
 i.e.,  $f(A) = B$



• bijection = inj + surj.

A topological space  
 is a pair  $X = (X, \tau)$   
 such that:

$\uparrow$  if  $\tau$  is understood  
 $\uparrow$  "the topology"  
 $\uparrow$  a subset of  $2^X$   
 $\uparrow$  a set

(1)  $\emptyset, X \in \tau$

(2) finite intersection property

$$\bigcap_{i=1}^n U_i \in \tau$$

(3) arbitrary union property

$$\bigcup_{i \in \alpha} U_i \in \tau$$

$\uparrow$  index set  
 $\uparrow$  can be inf.

|                                                                    |
|--------------------------------------------------------------------|
| elts of $\tau$<br>we call<br>"open sets"                           |
| for std topo in<br>$\mathbb{R}^n$ , open is open<br>as we know it! |



Let  $X$  be a set.  
We always have these  
two topologies:

① Discrete topology

- $\tau = 2^X$

- generated by the  
singleton sets  $\{\{x\} \mid x \in X\}$

② Indiscrete topology  
(trivial topology)

$$\tau = \{\emptyset, X\}$$

$A \in 2^X$  is closed  
iff  $X-A$  is open.

[Q] write the def. of  
topology using closed sets

$\tau$  is gen. by  $S$   
means:  $\tau$  is the  
smallest topology that  
contains  $S$ .  
AKA - start w/  $S$  + build all of  
 $\bigcap_{\text{finite}}, \bigcup_{\text{inf}}$  you can!



$\mathbb{R}$

• std topo (gen by  $\{(a,b) \mid a < b \in \mathbb{R}\}$ )

-  $(0,1]$  neither open nor closed

- ex of open:

$(0,1), (0,\pi) \cup (100,100.1)$

-  $[2, 4.02]$  is closed b/c  $(-\infty, 2) \cup (4.02, \infty)$  is open

• trivial topology.

- OPEN:  $\emptyset, \mathbb{R}$  nothing more!

-  $(0,1)$  neither open nor closed

• Discrete topology

- OPEN =  $2^{\mathbb{X}}$

ex:  $(0,1]$



$X, Y$  topo spaces

$f: X \rightarrow Y$  fun

$f$  is continuous if

$\forall$  open sets  $U \subseteq Y,$

$f^{-1}(U)$  is open in  $X$

"inverse of open is open"

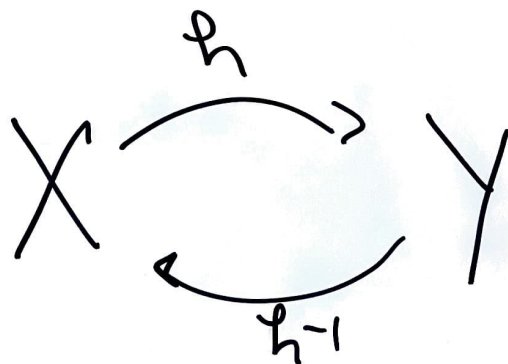
[Q] Prove this is equiv. to  $\epsilon$ - $\delta$  def.  
in the std topology on  $\mathbb{R}$

• Def: If  $\forall$  open  $A \subseteq X,$   $f(A)$  is open, we call  $f$  an open map.

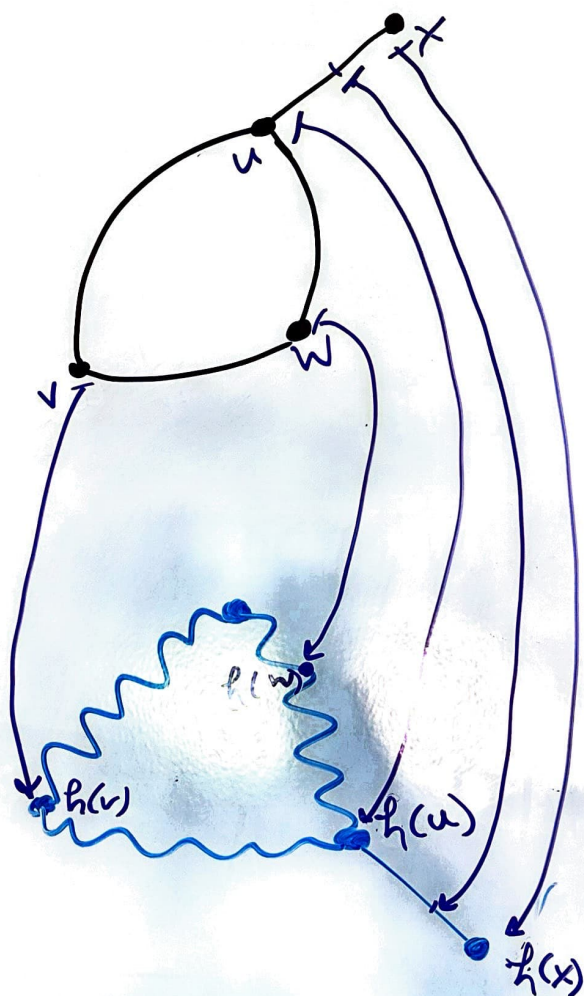
$X, Y$  topo spaces

The "strongest" notion of  
"sameness" is:

$X$  and  $Y$  are homeomorphic  
if  $\exists h: X \rightarrow Y$  that is a  
bicontinuous bijection.







(boundary of)  $\cong$  (boundary/surface of)  $\cong T \cong S^1 \times S^1$   
 coffee cup  $\cong$  donut  $\cong$  "the torus"

↗ see nice animation  
 on Wikipedia!



This is strict.

Instead, we look at equivalence,  
up to some property

- homotopy
- **homology**  $\neq$  **computable**

---

more restricted examples

- graphs, up to 3-colorability
- knots, up to twisting  $\#$

$G = (V, E)$  a graph <sup>embedded</sup> in  $\mathbb{R}^2$

geometric "measurable"

(specific to the embedding)

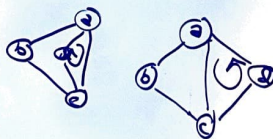
- lengths (MST's, etc.)
- orientation of verts.  
(3 verts: cw or ccw  $\Delta$ ?)
- exact locations of verts
- # of faces
- curvature
- # of  $\frac{?}{?}$  of verts

topological

(true  $\forall$  graphs homeomorphic to one we're investigating)

- # of connected components
- degree of a vertex
- graph density (how close to complete graph)
- # of cycles / loops
- the fact it is planar

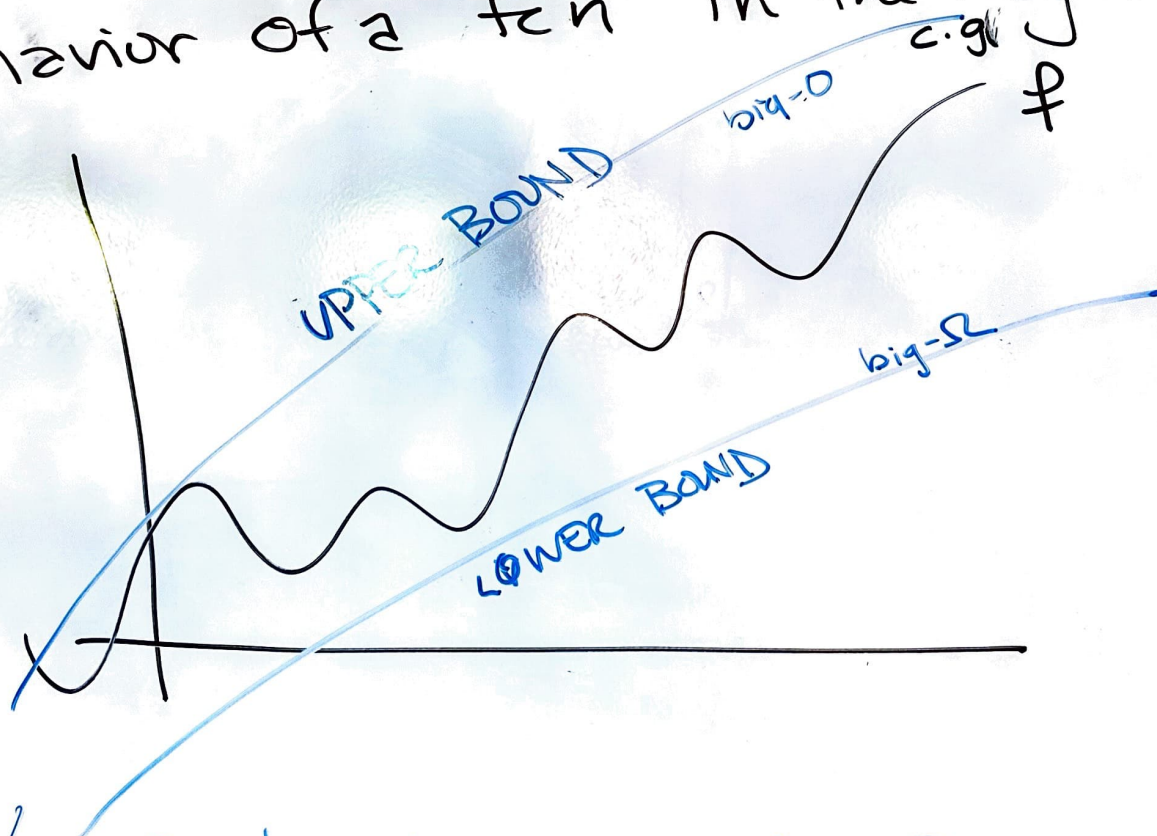
other examples outside a graph in  $\mathbb{R}^2$





# Asymptotics

How to describe the behavior of a fn "in the long run"



→  $\Theta$  = tight bound, when  $O = \Omega$

We say  
 $f: \mathbb{R} \rightarrow \mathbb{R}$  is  $O(g: \mathbb{R} \rightarrow \mathbb{R})$

when:

$\exists c > 0$  and  $n_0 \in \mathbb{N}$   
const. (scaling factor) (starting pt)  
such that

$\forall r \geq n_0, r \in \mathbb{R}$

$\uparrow \quad f(r) \leq c \cdot g(r)$

if this holds, we also say  
 $g(x)$  is  $\Omega(f(x))$